

Deelaka Jerome de Mel

Ja-Ela, Sri Lanka

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EDUCATION

University of Moratuwa

Sri Lanka

B.Sc. (Hons) in Engineering

2024 – Present

- Specializing in Electronic and Telecommunication Engineering.
- Current CGPA: 3.96 / 4.00

Coursework: Digital System Design, Digital Signal Processing, Electromagnetics, Electronic Circuit Design, Analog Circuit Design, Communication Systems Engineering, Electronic Control Systems

St. Joseph's College

Colombo 10

G.C.E. Advanced Level

2021 – 2023

- Physical Science Stream
- Z- Score : 2.65

KEY SKILLS

- Design & Simulation: Altium Designer, LTspice, MATLAB, Quartus
- Programming: Python, Java, C/C++, Verilog/SystemVerilog
- Core Competencies: Analog Circuit Design, PCB Design & Layout, Circuit simulation, RTL Design

ENGINEERING PROJECTS

Pure Sine Wave Inverter

📄 GitHub

- Designed and simulated a push-pull DC–DC converter to step up a 12 V input to a 400 V DC bus for a pure sine wave inverter system.
- Performed high-frequency transformer design, including core selection, turns-ratio calculation, and magnetic component sizing.
- Conducted circuit simulation and component selection for power-stage optimization and voltage regulation.
- Worked through each stage using LTspice for simulations and troubleshooting.

UART Implementation in FPGA

📄 GitHub

- Implemented a complete UART transceiver from scratch in Verilog HDL using discrete FSM logic (no IP cores), deployed on a Cyclone IV E FPGA, as part of the EN2111 Electronic Circuit Design module.
- Designed 4-state FSMs for both transmitter and receiver, operating at 9600 baud from a 50 MHz onboard clock via a `CLKS_PER_BIT` clock-divider parameter, with mid-period sampling on the RX side for noise immunity.
- Developed a top-level module integrating TX and RX, generating a 1-second tick to auto-increment transmitted data and driving received bytes to onboard LEDs and dual 7-segment displays.
- Verified functionality with a ModelSim loopback testbench, transmitting and validating byte-level data integrity with automated PASS/FAIL reporting.
- Confirmed hardware pin assignments and I/O mapping using Quartus Pin Planner for on-board testing.

Discrete Class-D Amplifier

📄 GitHub

- Designed and simulated a discrete-component Class-D audio amplifier operating at a 300 kHz switching frequency.
 - Performed component selection, circuit analysis, and LTspice-based verification of modulation, gate-drive, and output-stage performance.
 - Evaluated switching losses, efficiency, and output filtering to optimize amplifier operation.
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Analog Solar Tracker

 GitHub

- Designed a fully Analog Solar tracker
 - Utilized LTspice for a rigorous analysis of the power stages and PWM generation.
 - Implemented P and I control through analog components.
 - Designed a custom 4 layer PCB using Altium Designer.
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Configurable Integer Arithmetic & Matrix Multiplication Unit (Ongoing project)

EN3021 – Individual Project

- Designing configurable 8/16/32/64/128-bit integer multiplier and divider units in SystemVerilog, built entirely from scratch without vendor IP or libraries.
 - Investigating and comparing multiple hardware architectures (e.g. array, Wallace tree, Booth, Dadda for multiplication; restoring, non-restoring, SRT for division) to select designs optimized for latency versus logic resource utilization.
 - Implementing a configurable matrix multiplier supporting 3×3 , 5×5 , and 7×7 matrices, evaluated on throughput and area.
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Smart Bus Ticketing and Reservation System (SBTRS)

 GitHub

- Collaborated in a team to develop a scalable web-based bus ticketing and reservation platform using Spring Boot and Supabase.
 - Led the development of the Driver Module, implementing features for passenger manifest viewing, ticket verification, trip management, and role-based authentication.
 - Developed RESTful backend APIs and integrated the Driver Module with the centralized reservation and ticketing system to enable secure, real-time access to trip information.
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LEADERSHIP & INVOLVEMENT

IESL Student Chapter

University of Moratuwa

Active Member - Editorial Pillar

2025 – Present

- Managed technical documentation required for events organized by IESLSC.

St. Joseph's College

Colombo 10

Vice Captain - Basketball

- Led the team through intensive training programs and inter-school tournaments, emerging as champions at the U17 All island Basketball tournament.
- Coordinated team logistics and fostered a culture of discipline and strategic play among 15+ athletes (including non-team members).

AWARDS AND ACHIEVEMENTS

- St. Joseph's College, Colombo 10
 - Recipient of The Rev. Dr. B. Stanley Abeysekera Memorial Prize for the highest Z-score (2.65) at the GCE Advanced Level Examination; ranked 128th Island-wide.
 - Basketball Vice Captain: Led the school team to victory in multiple inter-school tournaments, managing team strategy and discipline. Arose as champions at the U-17 All Island School's Basketball championship.

- Member of the Science Club: Participated in inter-school science events, representing St. Joseph's College.

EXPERIENCE

Sampath Bank PLC

Hendala Branch

Intern

April 2023 – September 2023

- Processed financial documentation while ensuring accuracy and compliance with bank procedures.
- Worked with branch staff to support day-to-day banking operations and customer service.
- Developed professional communication, teamwork, and problem-solving skills in a fast-paced environment.

REFERENCES

Dr. Upeka Premaratne

M.E.Sc. (Western Ontario)

PhD (Melbourne)

LL.B. (OUSL)

Attorney-at-Law

Senior Lecturer - Grade 1

Department of Electronic and Telecommunication Engineering, University of Moratuwa

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